03-621 Week 12

Advanced Quantative Genetics

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Epigenetics and Epigenomics

Parts of Genes

Promoters

- Major on-off switches in genes
- Recruit transcription initiation complexes and RNA polymerase
- Most elements are upstream of the start site
- Some promoters lack these core elements

Enhancers

- Sequences that interact with activator proteins to boost transcription
- Frequently tissue-, stage-, or condition-specific
- Usually <20bp long
- Usually upstream of the promoter, but can be **far** upstream, downstream, or within a gene (in introns)

Insulator Elements

- Long-range looping can occur due to tissue-specific enhancers interacting long range
- Insulator elements prevent enhancer-promoter interactions
- Can restrict the activity of enhancers to defined chromatin domains (don't activate nearby genes).

Locus Control Regions

Can affect the transcription of multiple, distantly located genes, "choosing between them"

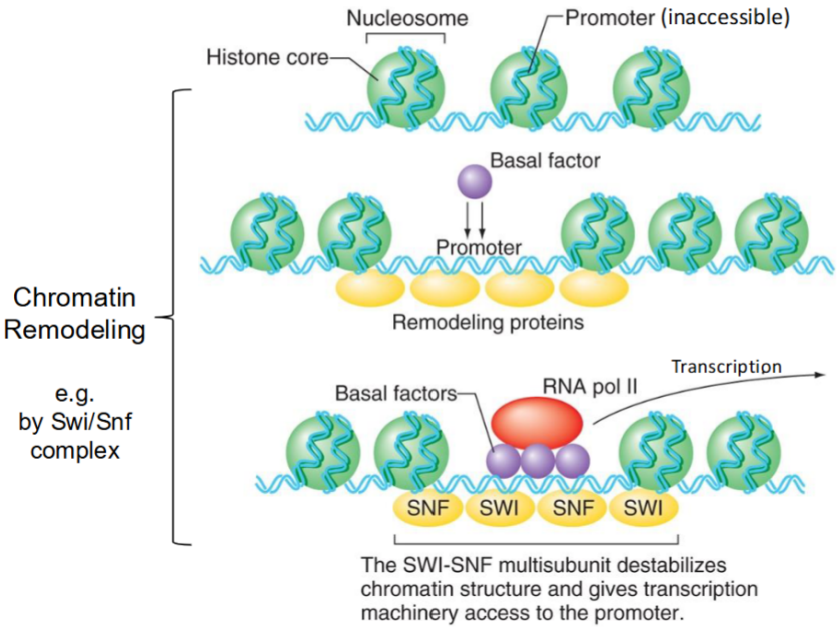
- For example, in the β-globin gene cluster, the same LCR can associate with different β-globin genes at different times in development, activating them sequentially.

Transcriptional Condensates: A New Regulatory Layer

- Transcriptional condensates are phase-separated clusters of transcription factors, mediator, and RNA Pol II
- Driven by intrinsically disordered regions (IDRs) and multivalent interactions
- Responsible for concentrating transcription machinery at active loci
- Enhancers nucleate condensates to regulate gene expression
- Highly dynamic: assmeble/disassemble in response to signals

Chromatin

- Chromatin compacts the DNA so it fits and remains organized in the nucleus.
- This structure also makes it hard for transcription factors and transcriptional machinery to access DNA



Epigenetics

- Regulation of Gene Expression at the level of chromatin structure, histone modification and DNA methylation
- ON/OFF epigenetic states can be established in response to:
  - Developmental program (cell type differentiation)
  - Environmental factors
- Epigenetic regulatory states (ON/OFF) can be inherited through somatic cell divisions
  - Provides a "cellular memory" of previous decisions in the cell lineage

Chromatin states arise from DNA and chromatin modifications.

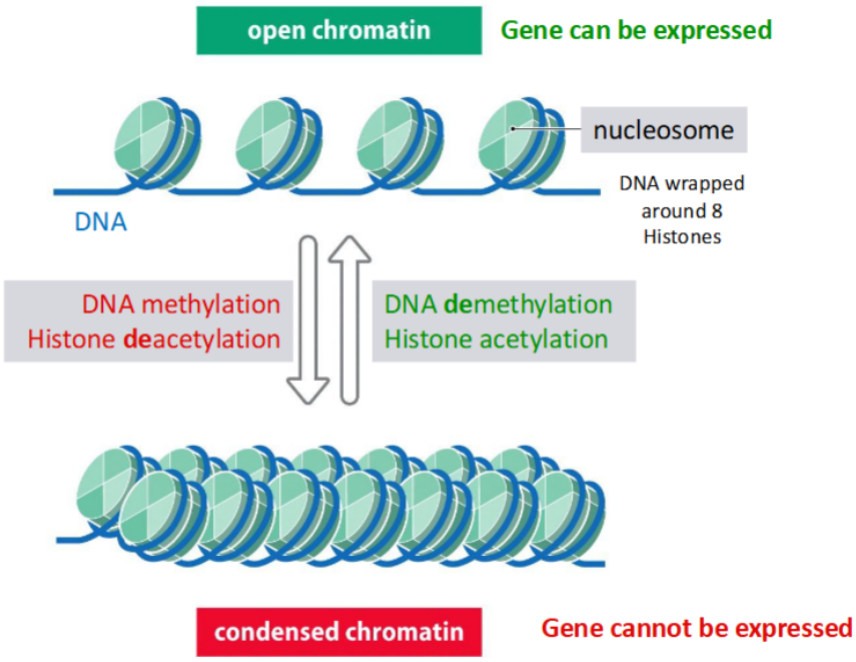


Figure 6.10 Genetics and Genomics in Medicine (© Garland Science 2015)

Different types of histone methylation are associated with open and condensed chromatin.

Contributions by DNA and Histone modification to different chromatin states

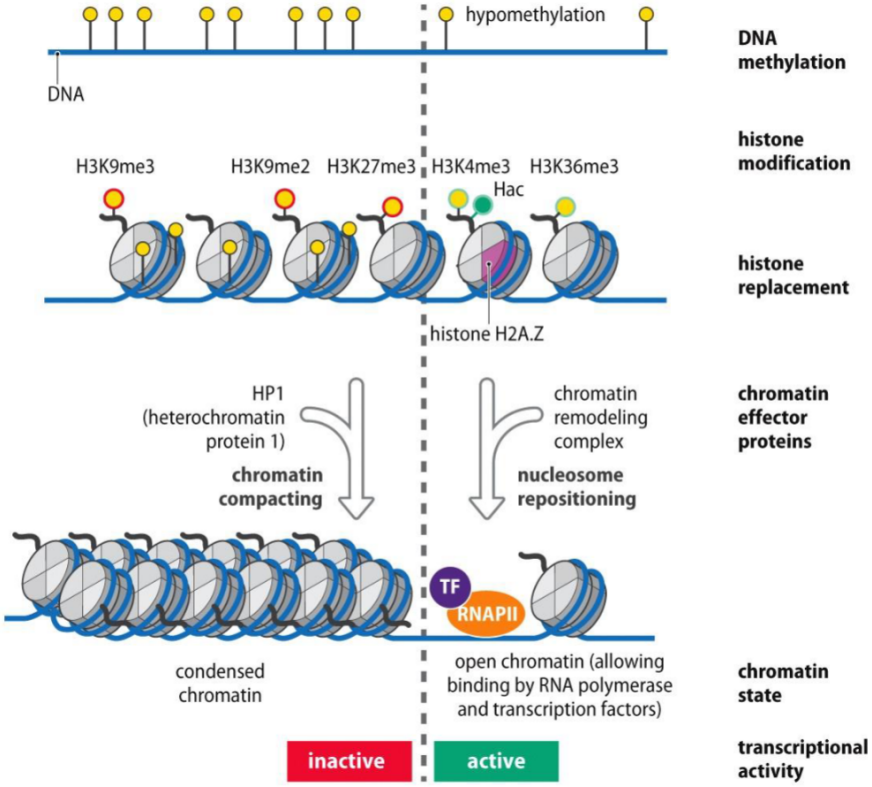


Figure 6.12 Genetics and Genomics in Medicine (© Garland Science 2015)

Histones H3 and H4 undergo many modifications on their positively charged N-terminal tails. H3 Lysines K9 and K27 can be:

- Methylated (inactive genes)
- Acetylated (active genes)

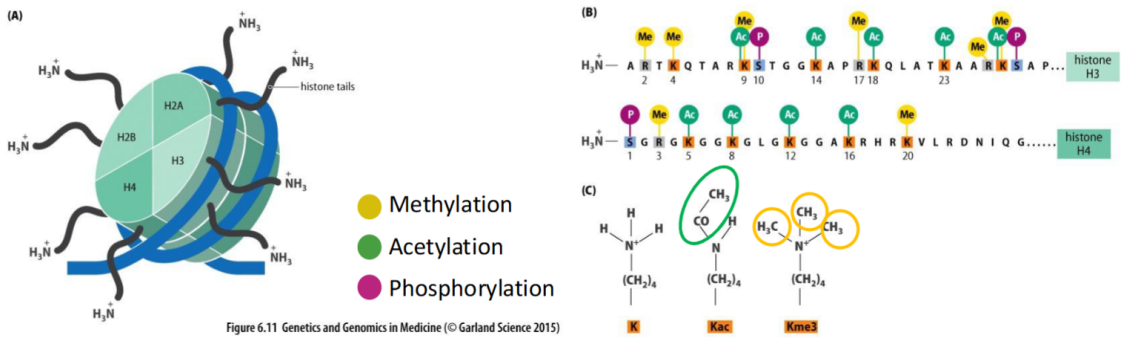
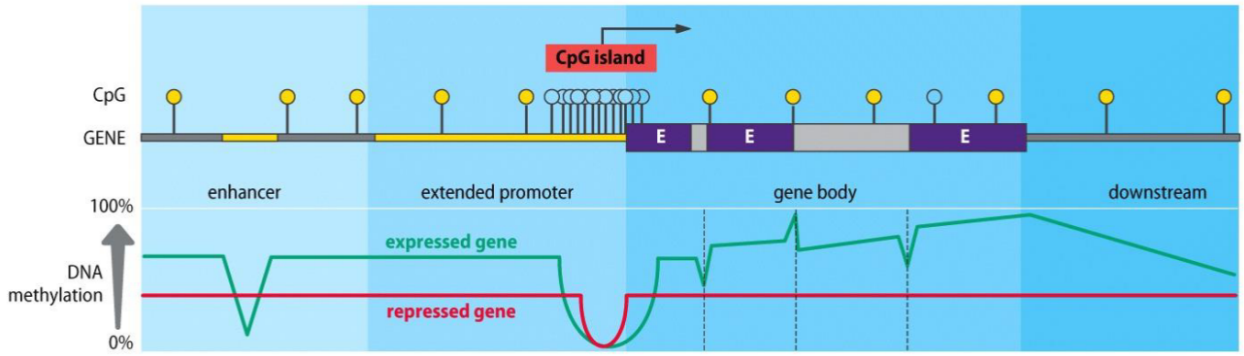


Figure 6.11 Genetics and Genomics in Medicine (© Garland Science 2015)

Examples of Histone Modifications Characteristic of different Chromatin States

Amino Acid	Euchromatin					Heterochromatin	
	Promoters		Enhancers		Gene Bodies	Facultative	Constitutive
	Active	Inactive	Active	Inactive	Inactive		
H3K4	H3K4me2, H3K4me3		H3K4me1, H3K4me2				
H3K9	H3K9ac	H3K9me3		H3K9me2, H3K9me3	H3K9me2, H3K9me3	H3K9me2	H3K9me3
H3K27	H3K27ac	H3K27me3	H3K27ac			H3K27me3	
H4K12	H4K12ac					H4K12ac	H4K12ac
H4K20							H4K20me3

CpG density and DNA methylation levels across a typical gene



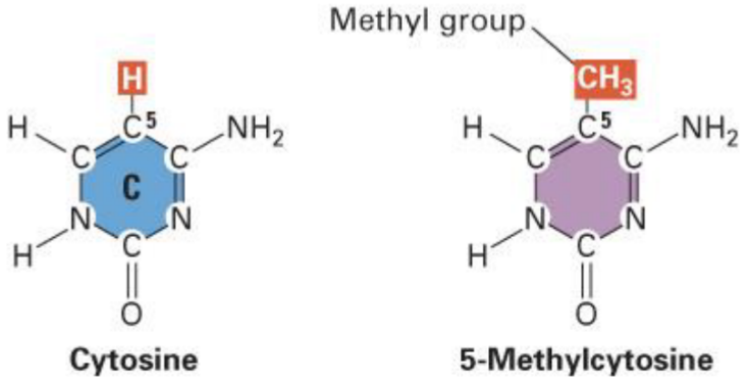
Box 6.3 Figure 1 Genetics and Genomics in Medicine (© Garland Science 2015)

CpG Islands:

- 300-3000 nucleotides long, with high G/C content.
- Methylation of C leads to silencing of gene expression
- Often found in transposons (suppresses hopping).
- 60-70% of CpGs are methylated in human somatic cells.

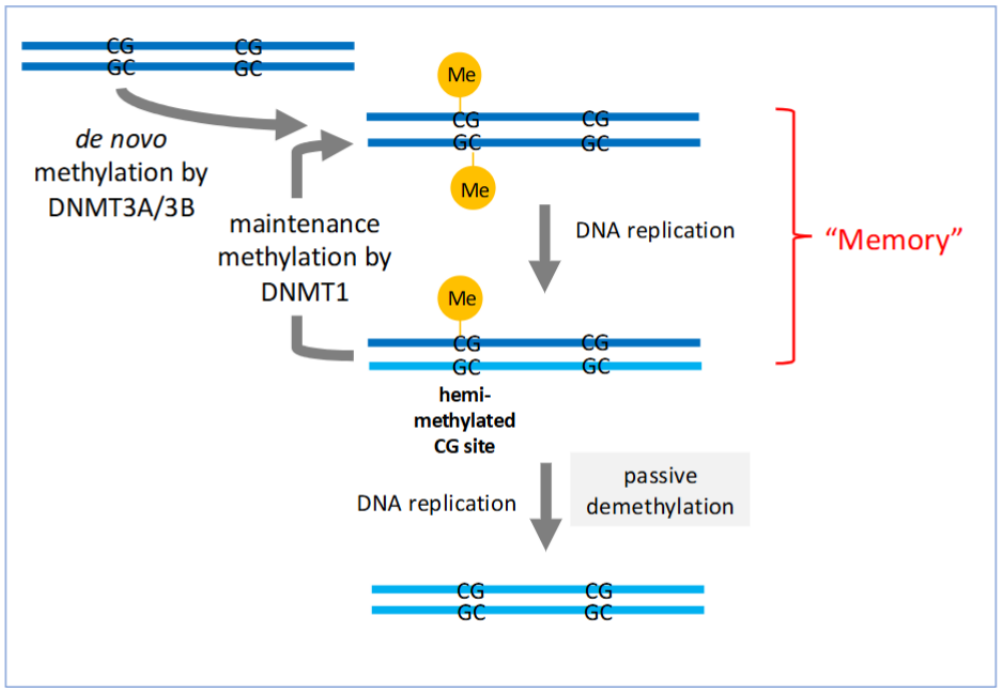
Epigenetics

- Heritable change in something "in addition to" or "on top of" DNA sequence.
- Methylation of C's in CpG islands by DNA Methyltransferase (DNMT)
- Results in local silencing of transcription
- Methylation is inherited through somatic cell divisions, through the action of "maintenance" methylases



DNA Methylation and Demethylation at CpG in Mammalian Cells

1. **De novo methylation** can establish a repressed state in response to developmental program or environmental factors
2. Methylation can be lost through subsequent rounds of cell division
3. **Maintenance methylation** maintains repressed state through somatic cell divisions. Accounts for cellular "memory" of past regulatory decisions



DNA Methylation in Neanderthals

We suspect that neanderthals diverged from humans through epigenetic evolution.

- Neanderthal genome sequenced in 2010
- 5-methyl-cytosine naturally deaminates to thymidine.
- Methylation sites "look" like mutations, compared to modern human genome sequences.
- The C $\rightarrow$ T "mutation" shows the methylation status in Neanderthals
- Over 99% of genome shows no methylation different from present-day humans

Comparing the HOXD genes in DNA of osteoblasts (bone cells)

- Neanderthals had *more* methylation at HOXD9 and HOXD10 genes.
- This may have led to *less* expression, accounting for differences in bone structure.
- LOF mutations in these genes lead to similar differences in mouse, human

Methylation Paradox

If methylation silences gene expression, and it is heritable through cell divisions, how can new progeny develop?

Answer: Methylation status is controlled dynamically

- Erased during development of germ cells
- Reset (where appropriate) during subsequent differentiation of gametes, fertilization, and development
- The repressed state is **reversible** in response to the developmental program (e.g., during gamete differentiation and early embryonic development, or in response to environmental factors)

Changes in DNA Methylation During Mammalian Development

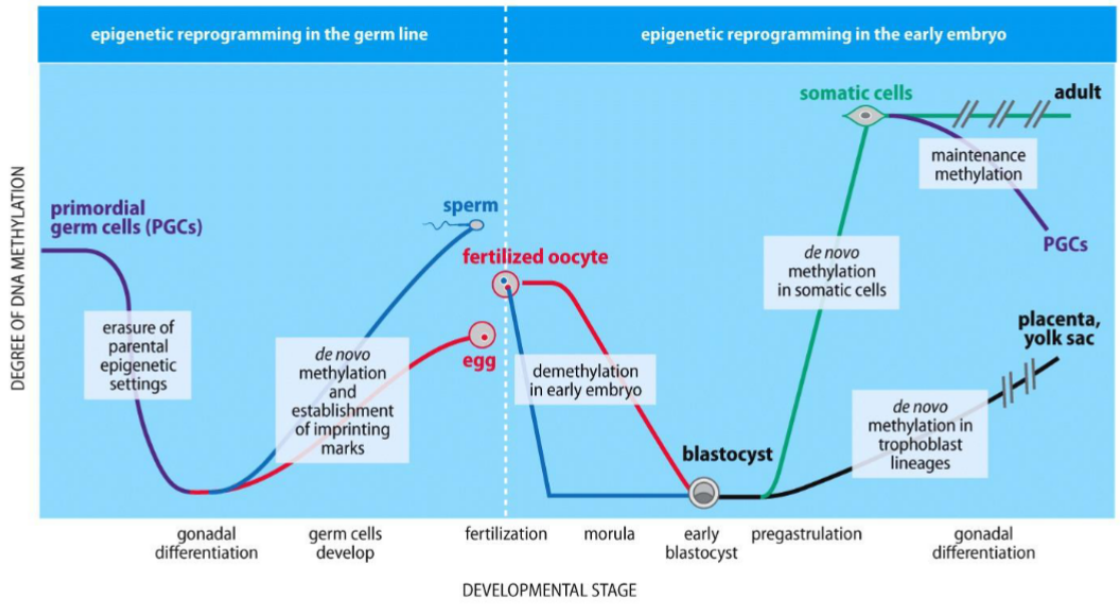


Figure 6.14 Genetics and Genomics in Medicine (© Garland Science 2015)

- CpG islands are demethylated in PGCs and in preimplantation embryos
- CpG islands become methylated during development for differentiation (as appropriate for sperm, oocytes, embryo, and adult somatic cells).
- Deficient DNA methylation is associated with Neural Tube Defects (folic acid vitamin supplement increases methylation - reduces risk)

Examples of Regulation by DNA Methylation During Development

Independent of parent of origin	**X-inactivation in somatic cells**	Random inactivation of most genes on either the paternal or maternal X
	Production of cell-specific Ig and T-cell receptors	Each mature B or T cell makes Ig or T-cell receptor chains using only one allele. Once a functional chain is made by gene rearrangement at one randomly selected allele, a feedback mechanism inhibits further rearrangements
	Production of cell-specific olfactory receptors	Each olfactory neuron expresses a single allele of just a single olfactory receptor (OR) gene (selected from several hundred OR genes) so that it fires in response to one specific odorant only. Depends on competition for a single monoallelic enhancer.
	Stochastic and environmentally induced	May be quite common? Intense interest in context of cancer, obesity, and diabetes.
Dependent on parent of origin	**X-inactivation in placenta**	Paternal X is always inactivated
	Genomic imprinting	Some genes are expressed only from the paternally inherited homolog and some only from the maternally inherited homolog. Examples: IGF2, H19, 15q11, etc.

Imprinting Occurs During Germ Cell Differentiation

- In mammals, we would not expect trans-generational inheritance of epigenetic changes acquired in response to environmental factors.
- But some specific epigenetic marks are transferred through the sperm or egg → **Imprinting**: parent-of-origin silencing / expression

Imprinting

- Imprinted genes are methylated differentially in maternally and paternally-derived chromosomes.
- Silenced loci remain off during development.
- Exhibited by several hundred human genes.
- Therefore, we only have one **active** copy of those genes. It is either the maternal or paternal copy, but it depends on the gene.
- The silenced allele can **switch** as it passes through a new germline.

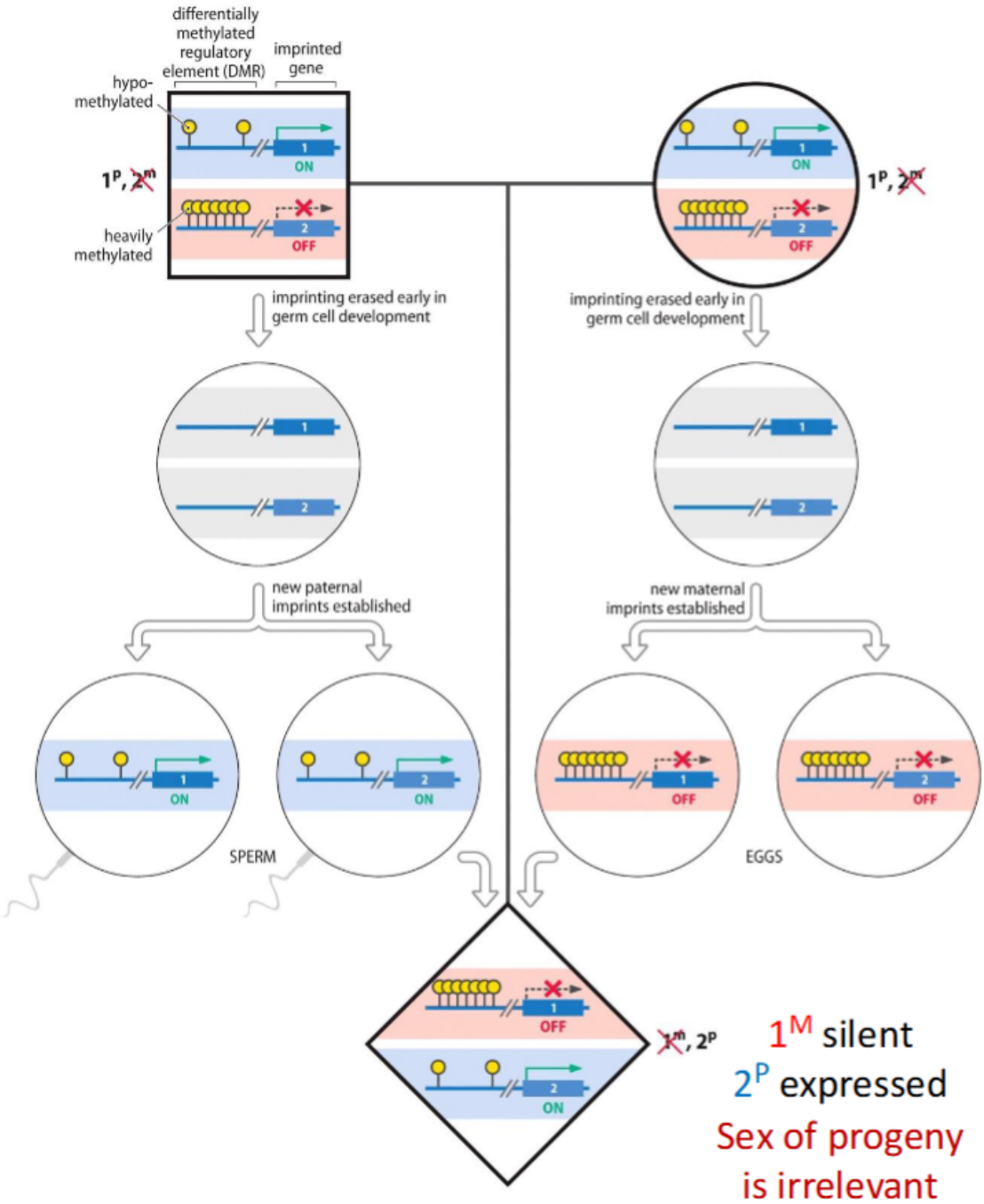
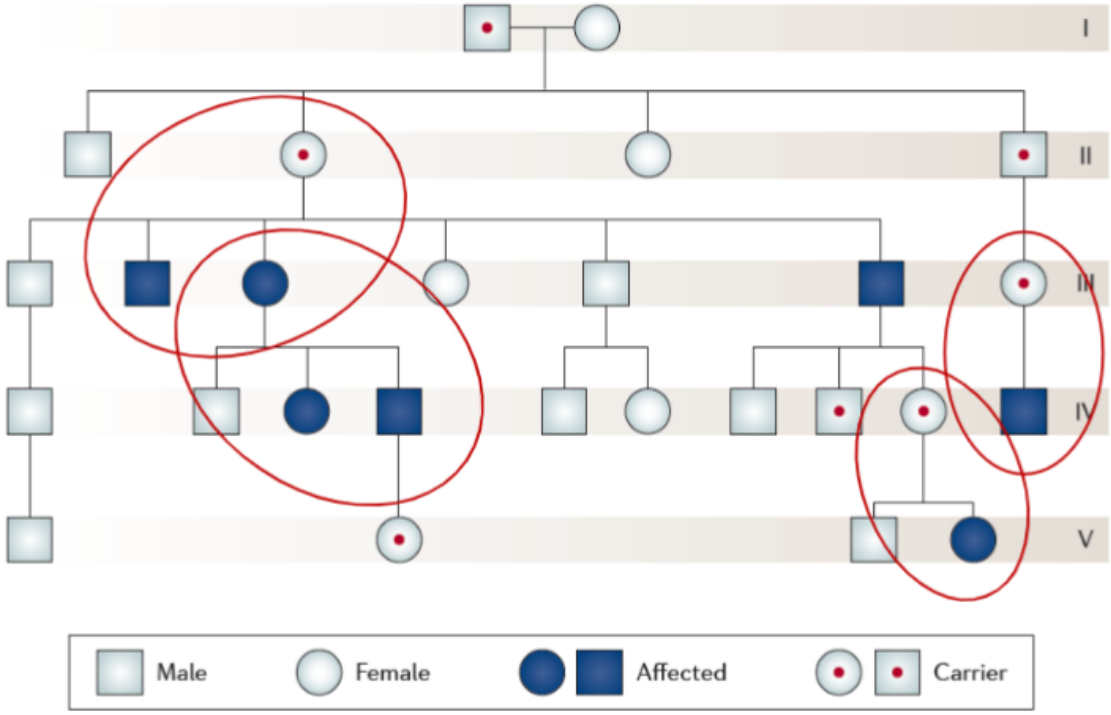


Figure 6.16 Genetics and Genomics in Medicine (© Garland Science 2015)

Some Phenotypic Traits Exhibit Imprinted Transmission



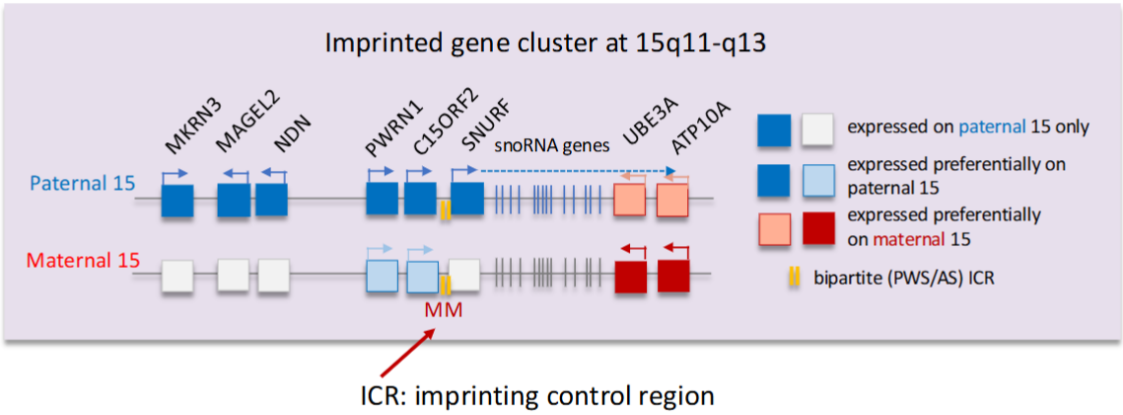
- **Parent-of-origin effects:** the disease phenotype associated with a DNA sequence variant is only observed when that variant is inherited from the parent of a particular sex (in this case, the female parent).

Pathogenesis Can Result from Underexpression *or* Overexpression of Imprinted Genes

Syndrome	Clinical Features	Molecular Basis of Imprinting Disorder	Cause	
			UPD*	Other
Pathogenesis due to under-expression of imprinted genes				
Prader-Willi	Developmental delay hypotonia, hyperphagia, obesity, heart disease	Lack of active alleles for imprinted snoRNA genes at 15q11.2	Mat. chr 15 $\sim 25\%$	Del pat. 15q11-13 $\sim 70\%$
Angelman	Developmental delay (severe) no speech epilepsy; ataxia	Lack of active allele for imprinted UBE3A gene at 15q11.2	Pat. chr 15 $\sim 25\%$	Del mat. 15q11-13 $\sim 75\%$
Pathogenesis due to over-expression of imprinted genes				
Beckwith-Wiedeman	Macrosomia/overgrowth; macroglossia umbilical defect	Biallelic expression of IGF2 at 11p15.5 (normally silenced on maternal 11) Biallelic expression of ncRNA that silences CDKN1C growth suppressor	Pat. 11	Loss of maternal ICR2** methylation $\sim 50\%$ Gain of maternal ICR1** methylation $\sim 5\%$ CDKN1C mutation $\sim 5\%$

Consider the causes of Prader-Willi and Angelman syndromes. They have the same causes (just in different parental chromosomes), but they have very different effects.

Different syndromes depending on which parent transmitted the deletion or UPD



- CpG methylation (M) silences transcription of **maternal** *SNURF* + *snoRNA*
- Run-on transcription of **paternal** *SNURF* + *snoRNA* interferes with transcription of *UBE3A* and *ATP10A*

As a result,

- Maternal UPD or Paternal deletion: No SNURF, snoRNA expression → Prader-Willi
- Paternal UPD or Maternal deletion: No UBE3A, ATP10 expression → Angelman

Epigenetic Mutations

Disruptions in the *regulation* of epigenetic silencing. They can result from:

- Mutations in the genes that encode the regulators of chromatin structure:
  - CpG islands
  - DNA methylases
  - Histone modifiers
  - etc.
- Environmental factors or stochastic events that change the activity of the regulators

Both types are called "mutations" because both can be inherited through **somatic** cell divisions

[insert 40, edited]

Traits Associated with Genetic Mutations of Chromatin Modifiers (i.e., "Secondary Epimutations")

[insert table, 41]

"Primary Epimutations"

Appear to play important roles in:

- cancer
- metabolic disease (e.g., Diabetes)
- behavioral traits (e.g., anxiety)

Are they heritable across generations?

Tools and Databases for Epigenetic / Epigenomic Analysis

[insert 42]

Functional Genomics

Genome-scale characterization of

- Chromatin structure
- Chromatin modifications
- Transcription factor DNA binding
- Protein interactions
- Transcript structures and amounts
- Regulatory interactions

These characterizations:

- complement linkage and GWAS data to identify disease genes and understand the effects of mutations
- help understand the functional organization of the genome and how it guides development and biological responses to the environment

Method to Identify Regions of Methylated DNA vs Unmethylated DNA

- Methylated: transcriptionally inactive
- Unmethylated: open for transcirption

Mapping of Methylated Cytosines: "Bisulfite Sequencing"

1. Chemistry:

   [insert 45]

2. Methods:

   [insert 46]

   Compare sequences to reference genome, or compare bisulfiite-treated vs untreated

   - Cs that were NOT changed to T were methylated

3. Applications: Identify changes in epigenetic state at specific genes

   - During development:
     - Different stages
     - Different cell types
     - Different tissues
     - Aging
   - In disease:
     - e.g., Some genes are inactivated epigenetically during cancer progression
   - During disease therapy:
     - e.g., effects of different types or dosages of drugs

4. Pros and Cons:
   - Pros:
     - CpG and non-CpG methylation throughout the genome is covered at single-base resolution
     - 5mC in dense, less dense, and repeat regions are covered
   - Cons:
     - Bisulfite converts unmethylated cytosines to thymidines, reducing sequence complexity, which can make it difficult to create alignemnts
     - SNPs where a cytosine is converted to thymidine will be missed upon bisulfite conversion
     - Bisulfite conversion does not distinguish between 5mC and 5hmC

Mapping of Methylated Cytosines: "Bisulfite sequencing"

5mC and 5hmC

[insert 50]

Oxidative Bisulfite Sequencing

[insert 51]

Differential Microarray Hybridization

Compares representation of specific DNA or RNA sequences in two samples

[insert 52]

Methods to Identify Regions of Condensed Chromatin vs Uncondensed Chromatin

- Condensed: Cannot be transcribed
- Uncondensed: Open for transcription

0.0.1 1. DNase Hypersensitive Site Mapping (DHSs or HSS)

Used to identify regions of decondensed ("open") chromatin corresponding to expressed genes or their regulatory elements

[insert 53]

ATAC-seq: Sequences only the "open" chromatin

[insert 54]

The Tn5 transposome simultaneously fragments the open chromatin DNA ($\sim$ 300 bp) and attaches priming site extensions to the fragments

DHSs and ATAC-seq libraries are sequenced to count the number of DNA fragments from each position

[insert 56]

Pros and Cons of DHSs and ATAC-seq

DHS:

- Pros:
  - Detects TF footprints
  - Great sensitivity at promoters
- Cons:
  - Time-consuming
  - DNase I sequence bias

ATAC-seq:

- Pros:
  - Simple, fast sample preparation
  - Low input
  - Detects TF footprints
  - Detects nucleosome occupancy
- Cons:
  - Fresh tissue isolation
  - Mitochondrial DNA contamination
  - Sequence bias of Tn5 transposase

Methods to Identify DNA Sequences which Specific Proteins are Bound

- Transcription regulatory factors
- Modified histones
  - Methylated
  - Acetylated
  - Etc.

This would tell us the genes that are potentially regulated by those factors (in particular tissues or cell types)

Chromatin Immunoprecipitation Sequencing (ChIP-Seq)

[insert 58]

Used to find binding sites for specific transcription factors.

1. Crosslink DNA and protein with formaldehyde to stabilize interactions
2. Digest chromatin with nuclease or sonication (limited digestion $\rightarrow$ 300 bp fragments)
3. Add antibody against the protein of interest
4. Capture and wash the antibody-protein-DNA complexes
5. Undo the crosslinks (heat) and sequence the co-purified DNA

ChIP-seq libraries are sequenced to count the number of DNA fragments (reads) from each genomic position

[insert 59]

0.0.2 Example of ChIP-seq data: FOXA3p binding at *APOA2* in Liver

[insert 64]

Combining ATAC-seq, HSS, and ChIPSeq Data

This allows us to detect occupation of an open chromatin site by a specific transcription factor

[insert 65]

CUT&RUn and CUT&TAG: ChIP alternatives

[insert 66]

Methods for Genome-wide Measure of Gene mRNA Expression

Genome-wide mRNA abundance measurements are made with microarrays or high-throughput RNA sequencing (RNAseq).

- Use to compare gene activity
  - In different cell types
  - In disease vs healthy cells
  - Under different conditions
    - Experimental treatments
    - Therapeutic treatments
    - Mutant vs. wild type

Measuring Gene Expression with RNA-seq

[insert 68]

Short read and long read sequencing

[insert 69]

The Encyclopedia of DNA Elements (ENCODE) Project

ENCODE Project estimates that 80% of the human genome is functional

[insert 70]